

**Laboratory Record
of Enterprise Application
Development Lab**

Roll No.: 160124737313

Exp No.: 9

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Experiment: Design and perform CRUD operations (Create, Read, Update, Delete) on a MongoDB Student database using mongosh.

Setup

Open terminal and start MongoDB server if not already running, then launch mongosh:

```
mongosh
```

```
C:\Users\JALAN>mongosh
```

```
Current Mongosh Log ID: 69e7bad3d69a310df73682d0
```

```
Connecting to:
```

```
mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
```

```
Using MongoDB: 8.2.7
```

```
Using Mongosh: 2.8.2
```

For mongosh info see: <https://www.mongodb.com/docs/mongosh-shell/>

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (<https://www.mongodb.com/legal/privacy-policy>).

You can opt-out by running the `disableTelemetry()` command.

The server generated these startup warnings when booting

```
2026-04-21T23:25:10.545+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
```

Create/use database:

```
test> use student
```

```
switched to db student
```

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A) CREATE – Insert Documents

Insert one document:

```
student>db.students.insertOne({  
  | rollNo: '22IT001', name: 'Swathi', branch: 'IT', year: 2,  
  | marks: { DS: 84, DBMS: 90 }, phone: '9876543210', skills: ['C', 'C++']  
  | })  
  | {  
    acknowledged: true,  
    insertedId: ObjectId('69e7c8ccd69a310df73682d1')  
  | }
```

Insert many documents:

```
student> db.students.insertMany([  
  | { rollNo: '22IT002', name: 'Shiva', branch: 'IT', year: 2, marks: { DS: 91, DBMS: 88 },  
    skills: ['JS', 'Node'] },  
  | { rollNo: '22IT003', name: 'Ravi', branch: 'CSE', year: 3, marks: { DS: 77, DBMS: 81 },  
    hostel: true },  
  | { rollNo: '22IT004', name: 'Kiran', branch: 'ECE', year: 1, marks: { DS: 73, DBMS: 79 },  
    attendance: 86 }  
  | ])  
  |  
  | {  
    acknowledged: true,  
    insertedIds: {  
      '0': ObjectId('69e7c8d8d69a310df73682d2'),  
      '1': ObjectId('69e7c8d8d69a310df73682d3'),  
      '2': ObjectId('69e7c8d8d69a310df73682d4')  
    }  
  | }
```

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Verifying count:

db.students.countDocuments()

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B) READ – Query Documents

Basic read:

student> db.students.find().pretty()

[

{

 _id: ObjectId('69e7c8ccd69a310df73682d1'),

 rollNo: '22IT001',

 name: 'Swathi',

 branch: 'IT',

 year: 2,

 marks: { DS: 84, DBMS: 90 },

 phone: '9876543210',

 skills: ['C', 'C++']

},

{

 _id: ObjectId('69e7c8d8d69a310df73682d2'),

 rollNo: '22IT002',

 name: 'Shiva',

 branch: 'IT',

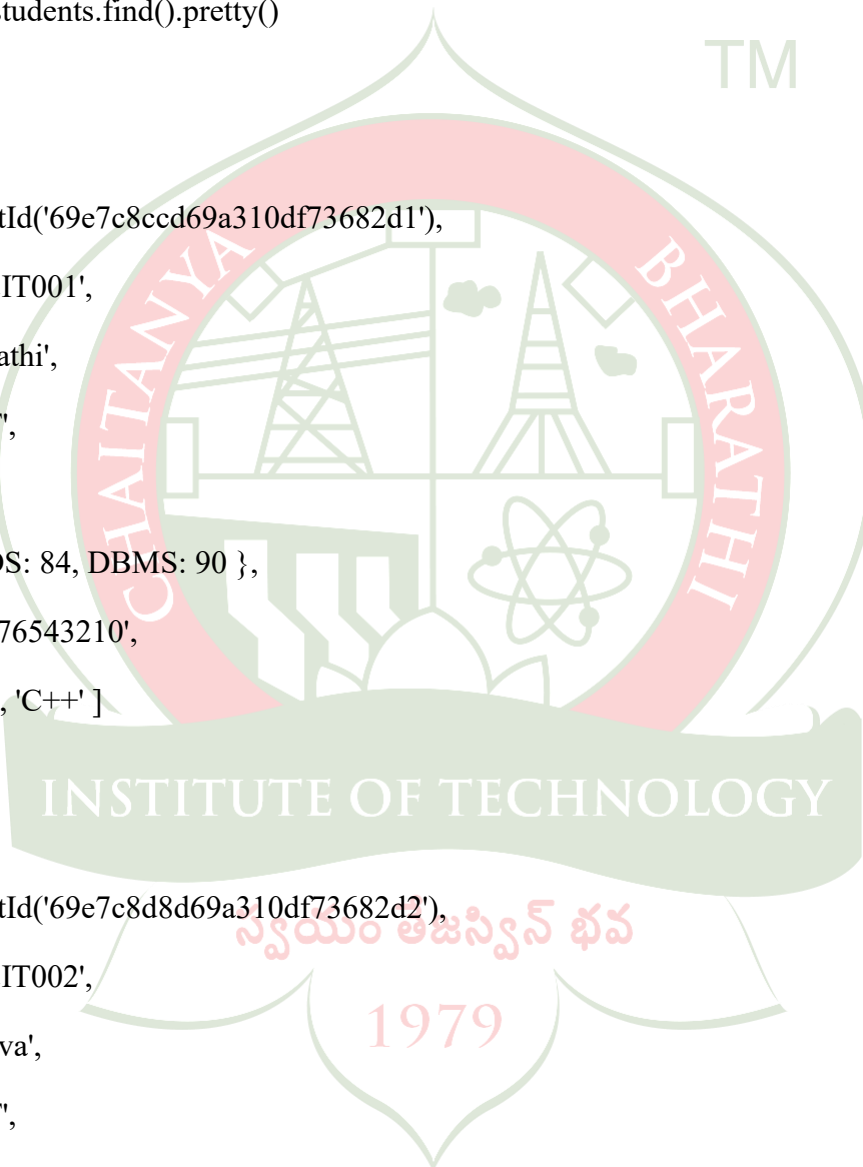
 year: 2,

 marks: { DS: 91, DBMS: 88 },

 skills: ['JS', 'Node']

},

{



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`_id: ObjectId('69e7c8d8d69a310df73682d3'),`

`rollNo: '22IT003',`

`name: 'Ravi',`

`branch: 'CSE',`

`year: 3,`

`marks: { DS: 77, DBMS: 81 },`

`hostel: true`

`},`

`{`

`_id: ObjectId('69e7c8d8d69a310df73682d4'),`

`rollNo: '22IT004',`

`name: 'Kiran',`

`branch: 'ECE',`

`year: 1,`

`marks: { DS: 73, DBMS: 79 },`

`attendance: 86`

`}`

`]`

Filter by field:

`student> db.students.find({ branch: 'IT' }).pretty()`

`[`

`{`

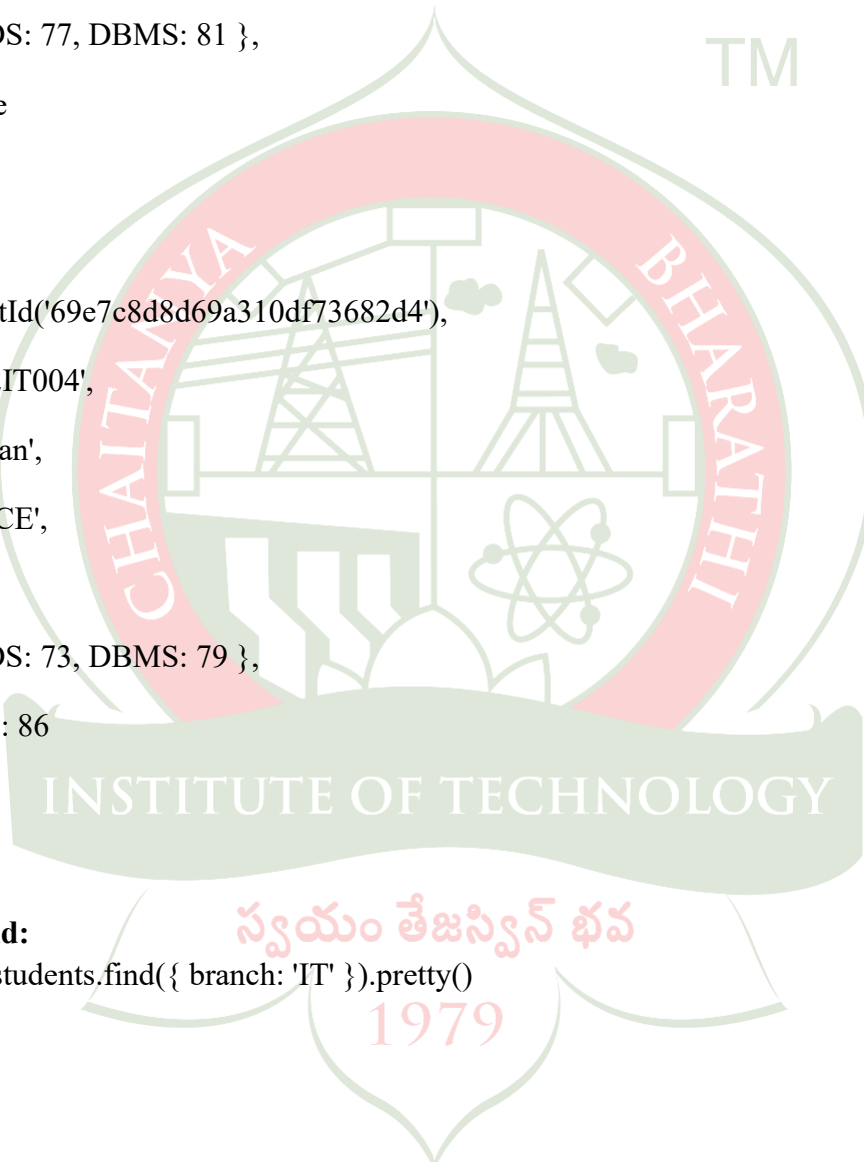
`_id: ObjectId('69e7c8ccd69a310df73682d1'),`

`rollNo: '22IT001',`

`name: 'Swathi',`

`branch: 'IT',`

`year: 2,`



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marks: { DS: 84, DBMS: 90 },

phone: '9876543210',

skills: ['C', 'C++']

},

{

_id: ObjectId('69e7c8d8d69a310df73682d2'),

rollNo: '22IT002',

name: 'Shiva',

branch: 'IT',

year: 2,

marks: { DS: 91, DBMS: 88 },

skills: ['JS', 'Node']

}

]

Filter with comparison operators:

student> db.students.find({ 'marks.DBMS': { \$gte: 85 } }).pretty()

[

{

_id: ObjectId('69e7c8ccd69a310df73682d1'),

rollNo: '22IT001',

name: 'Swathi',

branch: 'IT',

year: 2,

marks: { DS: 84, DBMS: 90 },

phone: '9876543210',

skills: ['C', 'C++']

},

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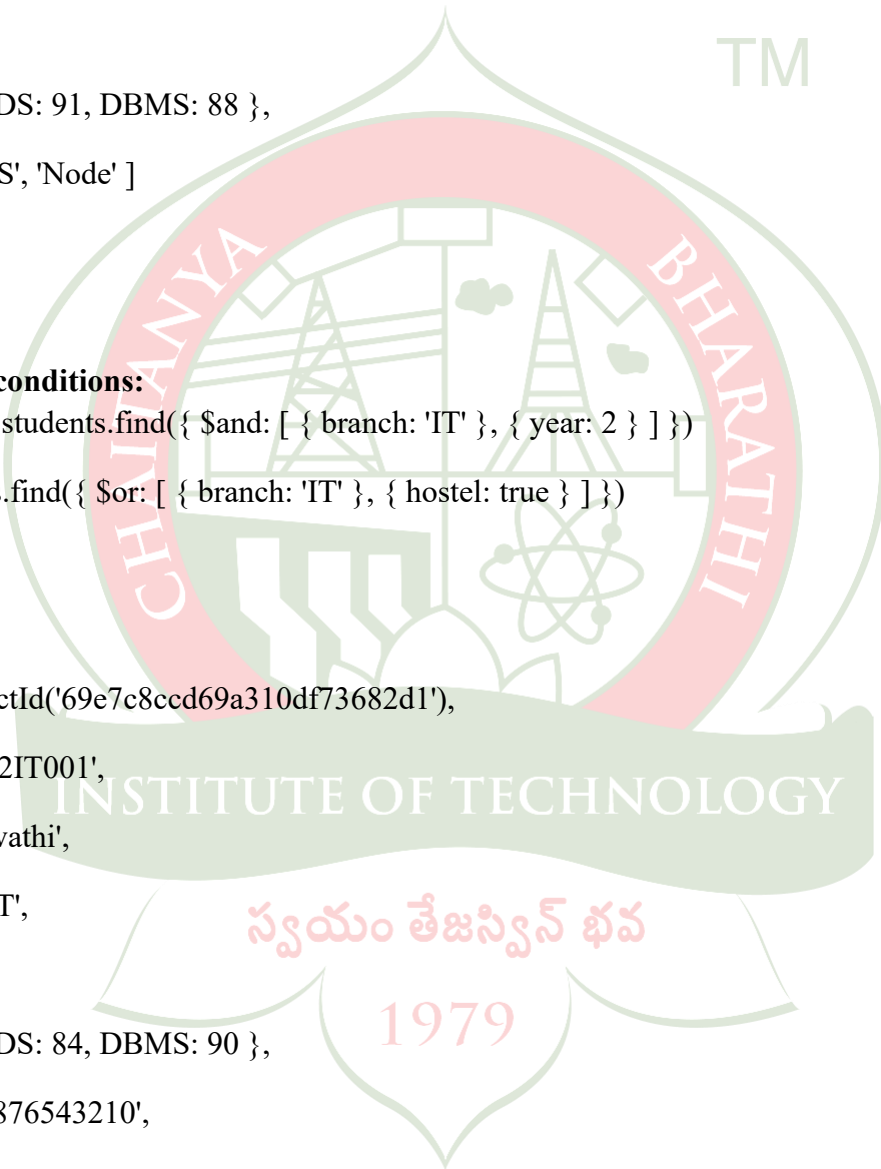
Date: _____

```
{  
  _id: ObjectId('69e7c8d8d69a310df73682d2'),  
  rollNo: '22IT002',  
  name: 'Shiva',  
  branch: 'IT',  
  year: 2,  
  marks: { DS: 91, DBMS: 88 },  
  skills: [ 'JS', 'Node' ]  
}
```

AND / OR conditions:

```
student> db.students.find({ $and: [ { branch: 'IT' }, { year: 2 } ] })  
| db.students.find({ $or: [ { branch: 'IT' }, { hostel: true } ] })
```

```
[  
  {  
    _id: ObjectId('69e7c8ccd69a310df73682d1'),  
    rollNo: '22IT001',  
    name: 'Swathi',  
    branch: 'IT',  
    year: 2,  
    marks: { DS: 84, DBMS: 90 },  
    phone: '9876543210',  
    skills: [ 'C', 'C++' ]  
  },  
  {  
    _id: ObjectId('69e7c8d8d69a310df73682d2'),  
    rollNo: '22IT002',
```



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```
name: 'Shiva',  
branch: 'IT',  
year: 2,  
marks: { DS: 91, DBMS: 88 },  
skills: [ 'JS', 'Node' ]
```

```
},
```

```
{
```

```
  _id: ObjectId('69e7c8d8d69a310df73682d3'),
```

```
  rollNo: '22IT003',
```

```
  name: 'Ravi',
```

```
  branch: 'CSE',
```

```
  year: 3,
```

```
  marks: { DS: 77, DBMS: 81 },
```

```
  hostel: true
```

```
}
```

```
]
```

Projection (select specific fields):

```
student> db.students.find({ branch: 'IT' }, { _id: 0, rollNo: 1, name: 1, 'marks.DBMS': 1 })
```

```
[
```

```
  { rollNo: '22IT001', name: 'Swathi', marks: { DBMS: 90 } },
```

```
  { rollNo: '22IT002', name: 'Shiva', marks: { DBMS: 88 } }
```

```
]
```

Sorting and limiting:

```
student> db.students.find({}, { _id: 0, rollNo: 1, name: 1, year: 1 }).sort({ year: -1, rollNo: 1  
}).limit(3)
```

```
[
```

```
  { rollNo: '22IT003', name: 'Ravi', year: 3 },
```

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```
{ rollNo: '22IT001', name: 'Swathi', year: 2 },
```

```
{ rollNo: '22IT002', name: 'Shiva', year: 2 }
```

```
]
```

Regex search (name starts with 'S'):

```
student> db.students.find({ name: { $regex: /^S/i } }, { _id: 0, name: 1, rollrollNo: 1 })
```

```
[
```

```
{ rollNo: '22IT001', name: 'Swathi' },
```

```
{ rollNo: '22IT002', name: 'Shiva' }
```

```
]
```

C) UPDATE – Modify Documents

Update one document using \$set:

```
student> db.students.updateOne(
```

```
| { rollNo: '22IT004' },
```

```
| { $set: { attendance: 90 } }
```

```
|)
```

```
{
```

```
  acknowledged: true,
```

```
  insertedId: null,
```

```
  matchedCount: 1,
```

```
  modifiedCount: 1,
```

```
  upsertedCount: 0
```

```
}
```

Add array values using \$addToSet (avoids duplicates):

```
student> db.students.updateOne(
```

```
| { rollNo: '22IT002' },
```

```
| { $addToSet: { skills: { $each: ['MongoDB', 'Express'] } } }
```


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|)

{

acknowledged: true,

insertedId: null,

matchedCount: 1,

modifiedCount: 1,

upsertedCount: 0

}

Remove a field using Sunset:

student> db.students.updateMany(

| { hostel: { \$exists: true } },

| { \$unset: { hostel: " } }

|)

{

acknowledged: true,

insertedId: null,

matchedCount: 1,

modifiedCount: 1,

upsertedCount: 0

}

Upsert (update if exists, else insert):

student> db.students.updateOne(

| { rollNo: '22IT005' },

| { \$set: { name: 'Meena', branch: 'IT', year: 1, marks: { DS: 68, DBMS: 72 } } },

| { upsert: true }

|)

{

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```
acknowledged: true,  
insertedId: ObjectId('69e7cd5b85c61965f6add7bf'),  
matchedCount: 0,  
modifiedCount: 0,  
upsertedCount: 1  
}
```

Replace an entire document (be careful – fields not provided are lost):

```
student> db.students.replaceOne(  
| { rollNo: '22IT003' },  
| { rollNo: '22IT003', name: 'Ravi Kumar', branch: 'CSE', year: 3, marks: { DS: 80, DBMS:  
83 } }  
| )  
|  
{  
acknowledged: true,  
insertedId: null,  
matchedCount: 1,  
modifiedCount: 1,  
upsertedCount: 0  
}
```

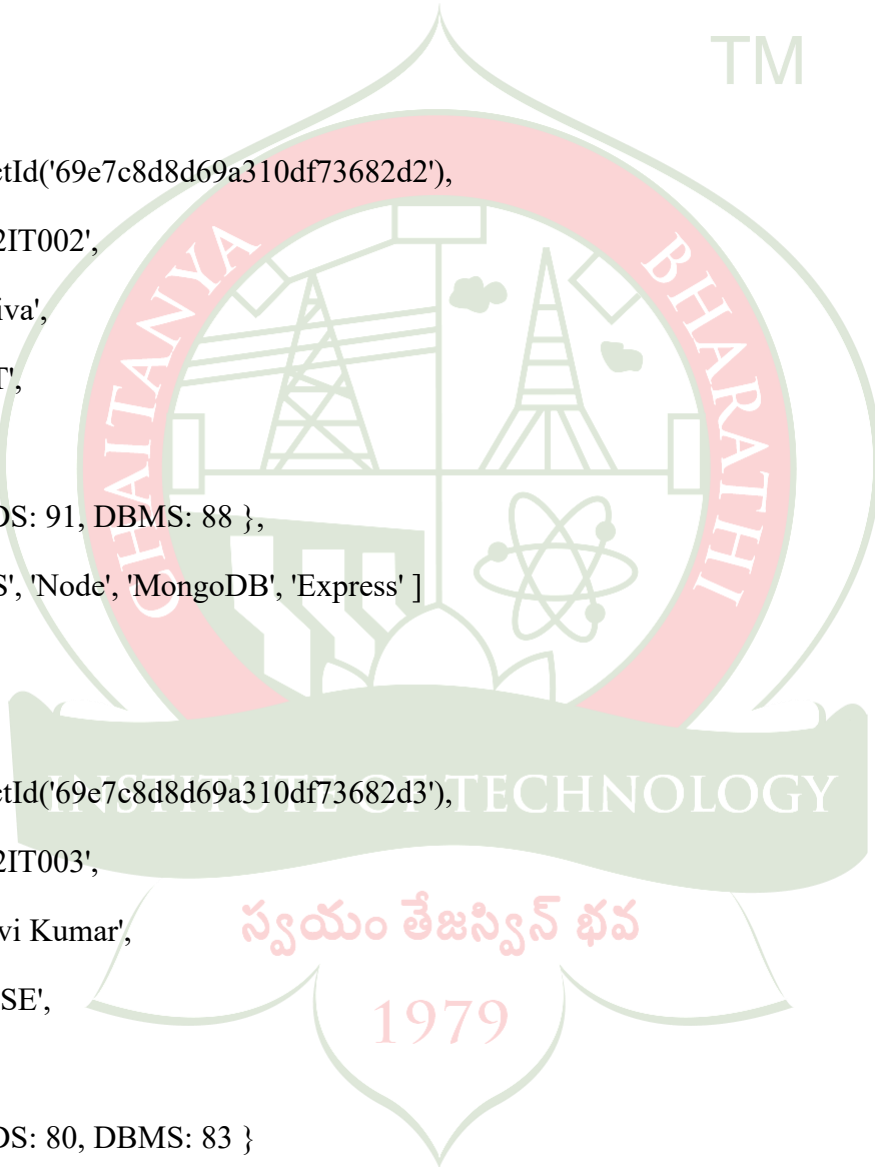
Showing updated data:

```
student> db.students.find().pretty()  
  
[  
  {  
    _id: ObjectId('69e7c8ccd69a310df73682d1'),  
    rollNo: '22IT001',  
    name: 'Swathi',
```

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```
branch: 'IT',  
  
year: 2,  
  
marks: { DS: 84, DBMS: 90 },  
  
phone: '9876543210',  
  
skills: [ 'C', 'C++' ]  
  
},  
  
{  
  
  _id: ObjectId('69e7c8d8d69a310df73682d2'),  
  
  rollNo: '22IT002',  
  
  name: 'Shiva',  
  
  branch: 'IT',  
  
  year: 2,  
  
  marks: { DS: 91, DBMS: 88 },  
  
  skills: [ 'JS', 'Node', 'MongoDB', 'Express' ]  
  
},  
  
{  
  
  _id: ObjectId('69e7c8d8d69a310df73682d3'),  
  
  rollNo: '22IT003',  
  
  name: 'Ravi Kumar',  
  
  branch: 'CSE',  
  
  year: 3,  
  
  marks: { DS: 80, DBMS: 83 }  
  
},  
  
{  
  
  _id: ObjectId('69e7c8d8d69a310df73682d4'),  
  
  rollNo: '22IT004',  
  
  name: 'Kiran',
```



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```
branch: 'ECE',  
  
year: 1,  
  
marks: { DS: 73, DBMS: 79 },  
  
attendance: 90  
  
},  
  
{  
  
  _id: ObjectId('69e7cd5b85c61965f6add7bf'),  
  
  rollNo: '22IT005',  
  
  branch: 'IT',  
  
  marks: { DS: 68, DBMS: 72 },  
  
  name: 'Meena',  
  
  year: 1  
  
}  
]
```

D) DELETE – Remove Documents

```
student> db.students.deleteOne({ rollNo: '22IT005' })
```

```
{ acknowledged: true, deletedCount: 1 }
```

Delete many documents with a filter (e.g., year < 2):

```
student> db.students.deleteMany({ year: { $lt: 2 } })
```

```
{ acknowledged: true, deletedCount: 1 }
```

Show remaining documents:

```
student> db.students.find({}, { _id: 0, rollNo: 1, name: 1, branch: 1, year: 1 }).sort({ rollNo:  
1 })
```

```
[
```

```
{ rollNo: '22IT001', name: 'Swathi', branch: 'IT', year: 2 },
```

```
{ rollNo: '22IT002', name: 'Shiva', branch: 'IT', year: 2 },
```

```
{ rollNo: '22IT003', name: 'Ravi Kumar', branch: 'CSE', year: 3 } ]
```

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E) Useful Extras (Optional but Recommended)

Create unique index on rollNo to prevent duplicates:

```
student> db.students.createIndex({ rollNo: 1 }, { unique: true })
```

rollNo_1

Count and distinct queries:

```
student> db.students.countDocuments({ branch: 'IT' })
```

```
| db.students.distinct('branch')
```

```
[ 'CSE', 'IT' ]
```

Aggregation: average DBMS marks by branch:

```
student> db.students.aggregate([
```

```
| { $group: { _id: '$branch', avgDBMS: { $avg: '$marks.DBMS' } } },
```

```
| { $sort: { avgDBMS: -1 } } ]
```

```
| )
```

```
[ { _id: 'IT', avgDBMS: 89 }, { _id: 'CSE', avgDBMS: 83 } ]
```

